MIL-P-21769A
31 January 1969
SUPERSEDING
MIL-P-21769
15 January 1962

MILITARY SPECIFICATION

PLUGS, FUSIBLE, REFRIGERANT-12

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers fusible plugs designed to function as thermal relief safety devices in refrigeration systems using Refrigerant-12.

1.2 Classification.- Fusible plugs shall be of the following sizes, as specified (see 6.2).

Size 1 - Melting point range of 157 F. through 170 F.

Size 2 - Melting point range of 208 F. through 220 F.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

* FEDERAL

QQ-B-626 - Brass, Leaded and Non leaded: Rod, Shaped, Forgings, and Flat Products With Finished Edges (Bar and Strip).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners or Pallet-Type Base.

FSC 5340

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MIL-STD-129 - Marking for Shipment and Storage.
MS 16993 - Plugs, Fusible, Refrigerant-12.

(Copies of specifications and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

UNDERWRITERS' LABORATORIES INC. (UL)

Standard No. 465 - Air Conditioners, Central Cooling.

(Copies of UL publications may be obtained from the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; Walt Whitman Road, Melville, Long Island, N.Y. 11746, or 1655 Scott Boulevard, Santa Clara, California 95050).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification B 16 - Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill. 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P. Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a sample of fusible plugs covered by this specification shall be submitted or made available to the Contracting Officer or his authorized representative for approval. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.2 Standard product.- The fusible plugs shall be the standard product of the manufacturer except for modifications required by this specification.

3.3 UL Standard.- Fusible plugs shall be tested and approved under the applicable requirements of UL Standard No. 465, Air Conditioners, Central Cooling.

3.3.1 UL certification.- Prior to award of the contract, the bidder shall submit, at the request of the Contracting Officer, satisfactory evidence that the fusible plugs he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 465. Acceptable evidence of meeting the applicable requirements of UL Standard No. 465 shall be the UL label, an approved UL listing, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the fusible plugs offered have been tested and conform to the applicable requirements of UL Standard No. 465.

3.4 Materials.- Materials that are not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.4.1 Brass.- Brass bar stock shall conform to copper alloy 360, half hard temper of QQ-B-626 or ASTM B 16.

3.5 Design.- The design of the fusible plugs shall conform to the requirements of MS 16993.

3.6 Finish.- The fusible plugs shall have the manufacturer's standard commercial finish.

3.7 Marking for identification.- In addition to the fusible metal melting point indicated on MS 16993, the manufacturer's name or equally identifiable trade name shall be clearly applied to the hexagon body of the fusible plug.

3.8 Workmanship.- The fusible plugs shall be free of burrs, rough edges, and disfigured threads.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects listed in table I and for dimensions specified.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 End item inspection.- The lot shall consist of all the plugs of one size offered for inspection at one time. The sample unit shall be one completely fabricated plug.

* 4.3.2.1 Visual examination.- Examination shall be in accordance with table I. The inspection level shall be level II with acceptable quality levels (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished as required.		X
Design	Any characteristic not in accordance with the specified requirements.	X	
Construction and workmanship, general (applicable to all components and materials)	Fractured, split, bowed, malformed, or otherwise impaired.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel.	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly	Not properly assembled or secured.	X	
Marking identification	Missing, incomplete, not legible.		X

* 4.3.2.2 Dimensional examination.- Inspection shall be made of the fusible plugs to determine compliance with dimensions specified on MS 16993. Any dimension not within specified tolerances shall be classified a defect. The inspection level shall be I with an AQL of 6.5, expressed in terms of defects per hundred units.

* 4.3.2.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as outlined in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examination	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged or otherwise defective affecting serviceability.
Weight of contents	Net weight exceeds requirements. Contents more than required.

* 4.3.2.3.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.3.3 Code and standard compliance.- Proof of compliance with the requirements of 3.3 shall be made available to the Government.

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* 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Twenty-five plugs, of one type only, shall be packaged in a snug-fitting paperboard box conforming to variety 1, style III or IV, type B, class 2 of PPP-B-566 or type I, variety 1, class A, style 1 of PPP-B-676. Prior to packaging, the flare connection end of each plug shall be provided with a snug-fitting paperboard sleeve or plastic cap. The paperboard sleeve shall abut the shoulder of the hexagonal body and extend approximately one-fourth of an inch beyond the flare connection end. The plugs shall then be placed in the box with the pipe thread end facing up, five long, five wide, and one deep. Box closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C.- The plugs shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Plugs of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2 Level B.- Plugs of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2), the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Plugs, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

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5.3 Palletization.- Unless otherwise specified (see 6.2), plugs packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The fusible plugs covered by this specification are for use with refrigerant receivers covered by MS 17291.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size (MS number) of fusible plug required (see 1.2).
(c) First article.- When a preproduction sample is required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) If palletization is not required (see 5.3).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - GL
Navy - YD
Air Force - 82

Review activities:
Navy - YD
Air Force - 82, 85

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Preparing activity:

Army - GL

Project No. 5340-0428

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Plugs, Fusible, Refrigerant-12 MIL-P-21769A

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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